Evapotranspiration and Yield Responses of Cotton to Irrigation

Abstract

This study evaluates the effects of varying irrigation schedules on cotton yield and water use efficiency in semi-arid loamy soils. We present updated crop coefficients and soil moisture thresholds.

1. Introduction

Precision irrigation requires accurate evapotranspiration (ETc) modeling over different growth stages. Loamy soils, common in cotton-growing regions, have a volumetric field capacity extending to 25%.

2. Evapotranspiration Calculation

We calculate ETc using the dual crop coefficient approach as follows:

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ETc = (Kcb + Ke) * ETo
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Where ETo is the reference evapotranspiration, Kcb is the basal crop coefficient, and Ke is the soil evaporation coefficient.

3. Growth Stage Parameters

The optimal moisture limits and coefficients for each growth stage are detailed in Table 1.

Growth Stage	Duration (Days)	Kc Value	Min Moisture (%)	Max Moisture (%)
Germination	15	0.40	18	25
Vegetative	45	0.75	16	23
Flowering	30	1.15	15	22
Boll Formation	40	1.05	14	20
Maturity	20	0.60	12	18

4. Conclusion

Water stress during the flowering stage leads to a 30% reduction in boll retention. Maintaining soil moisture above 15% is critical during this period. The wilting point for cotton in these loamy soils is 12%.